

Jerry Wang

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Education

Master of Science in Applied Machine Learning

University of Maryland - College Park

May 2026

College Park, MD

Bachelor of Science in Computer Science

Rutgers, the State University of New Jersey - New Brunswick

May 2023

New Brunswick, NJ

Skills

Languages: Python, C++, C, Java, SQL, Swift, Typescript, React

AI/ML: PyTorch, TorchVision, TensorFlow, Scikit-learn, YOLO, OpenCV, Reinforcement Learning, CUDA

Cloud & Infrastructure: AWS (S3, EC2, Lambda, CloudWatch), Docker, Kubernetes, Azure, FastAPI

MLOps & Tools: CI/CD, Git, Linux/macOS/iOS, pandas, NumPy, SciPy, Jupyter, ROS

Experience

Software Engineer Intern | Eslite

Oct 2020 – Aug 2021

- Built a distributed, reproducible data-validation workflow with unit, integration, and regression pipelines in CI/CD that caught regressions before release and lowered the post-release defect rate
- Redesigned an AWS pipeline – restructuring S3 storage and replacing synchronous Lambda processing with event-driven messaging in production – cutting data-retrieval latency by 30%
- Designed automated CloudWatch monitoring and alerting playbooks, adopted across the engineering team, that surfaced failure conditions earlier and reduced recurring server outages

Software Engineer Intern | Acer

Jun 2019 – Aug 2019

- Designed and trained a CNN and shipped it through Azure CI/CD to production line for automated visual inspection, achieving 95% defect-classification accuracy across 100,000+ image samples (held-out validation)
- Optimized and refactored on-device inference under runtime constraints, deploying a multithreaded model on embedded ARM/Raspberry Pi to cut AGV navigation errors by 35% (measured against weekly fault logs)
- Drove ML adoption across 30+ engineers by translating model architectures and trade-offs into documentation and workshops for technical and non-technical stakeholders

Projects

Personal AI Agent (On-Device LLM Inference) | MLX, Apple Silicon, FastAPI, Swift/iOS, Llama

Jun 2026 – Present

- Built a personal assistant running LLM inference entirely on local hardware – a self-hosted MLX inference server on Apple Silicon with a SwiftUI iPhone streaming client – keeping all user data on-device with no third-party API calls
- Implemented agentic tool use (web search, calendar, and reminders) with structured tool-calling, letting the assistant take real actions rather than only generating text
- Designed a human-in-the-loop confirmation layer that gates all state-changing (write) actions behind explicit user approval, preventing unintended side effects from autonomous tool calls

Depth-Based Autonomous Obstacle Avoidance for UAVs | RL, PPO, Gazebo, ROS, PX4, HPC

Sept 2025 – Dec 2025

- Designed and trained the perception-to-control RL policy (PPO) mapping stereo-depth streams to continuous velocity commands, validated across 50+ automated closed-loop trials on a PX4/Gazebo software-in-the-loop (SITL) bench
- Built a reproducible evaluation harness over a modular ROS2 node graph that decoupled perception, planning, and control into independently testable services – surfacing brittle transfer cases early and giving the team a shared regression baseline to iterate the policy against held-out environments

CARLA Autonomous Driving Perception Pipeline | PyTorch, OpenCV, YOLO, CARLA, C++/CUDA

Jun 2025 – Sept 2025

- Independently designed and built an end-to-end driving-perception pipeline over four months, engineering a reproducible flow spanning image ingestion, preprocessing, augmentation, and automated label resolution over a multimodal labeled dataset – improving scenario coverage and dataset quality
- Optimized the real-time path, porting critical preprocessing and inference stages from Python to modular, object-oriented C++/CUDA to cut per-frame inference latency by 25% (measured by GPU profiling) and sustain real-time throughput

Certification

Building Toward Computer Use with Anthropic

Aug 2025

NVIDIA AI Infrastructure and Operations Fundamentals

Jul 2025